# Amine Benmansour

**Application architect · Java, Kafka, applied AI · cards, payments and fraud**

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## PROFILE

14 years of Java experience, 11 of them inside the systems of major French banks (Crédit Agricole, Société Générale and BPCE groups, and Bpifrance). I designed the application and data-flow architecture of card, mobile-payment and fraud-prevention systems there, and led their delivery, most often as tech lead. I apply generative AI to the work of engineering teams: a production incident-analysis agent, agent-assisted CVE remediation, and AI assistance that runs locally.

## SKILLS

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|---|---|
| **Architecture** | application and data-flow architecture, microservices, event-driven architecture, DDD, hexagonal architecture, Event Sourcing / CQRS, sagas and compensation, idempotency, REST / OpenAPI, ADRs |
| **Applied AI** | coding agents, MCP servers (code repository, tickets, documentation, logs, image registry), GitHub Copilot, Claude Code |
| **Card payments** | issuing and card lifecycle (activation, blocking, PIN), Apple Pay / Google Pay tokenisation, balance check and Visa response-time constraints, fraud, disputes |
| **Flows and data** | Kafka (Avro contracts, Schema Registry, retry, dead-letter), IBM MQ, RabbitMQ, ActiveMQ, Spring Batch, CFT, PostgreSQL, Oracle, MongoDB, SQL Server, Flyway |
| **Integration** | core banking, card processor, IBM mainframe (CICS / CTG), OAuth 2.0, SAML |
| **Platform** | Kubernetes (GKE, OpenShift), Google Cloud, Helm, Kustomize, ArgoCD, GitLab CI/CD, Jenkins, SonarQube |
| **Operations** | Dynatrace, Grafana, Elasticsearch / Kibana, Datadog, Gatling load testing, production incident management |
| **Development** | Java 8 to 21, Spring Boot 2 and 3, GraalVM, TDD / BDD (Cucumber), JUnit, Testcontainers |
| **Regulation** | PCI DSS, DORA, AML (KYC), ACPR requirements |

## EXPERIENCE

### LCL (Crédit Agricole group)

December 2025 – present

*Senior Java engineer, application and data-flow architecture*

- Customer fraud-confirmation journey (card, transfer, cheque): design and development of the Java 21 / Spring Boot 3 microservices, the Kafka flows with Avro contracts, the REST APIs and the mainframe exchanges (CICS / CTG), and later enhancements.
- Error recovery without duplicate effects: Kafka retry strategy, replayable dead-letter topic and a step-tracking table. Each successful step is recorded and never replayed, so a replay never blocks a card twice; alerting on failures.
- Resilience pattern documented and presented to other teams for Kafka consumers that chain several external calls: orchestration, PostgreSQL step-tracking table, dead-letter and monitoring.
- Applied AI: incident-analysis agent run on demand by a business analyst or a developer, with the AI tool approved by the client (logs, tickets, monitoring and deployed images, through MCP servers); presented to other teams. Agent-assisted CVE remediation.
- Secure file-transfer component: Spring Batch, encryption, CFT.

*Java 21, Spring Boot 3, Kafka, Avro, REST, CICS / CTG, PostgreSQL, Spring Batch, CFT, Kubernetes, ArgoCD, GitLab CI, Dynatrace*

### BforBank (Crédit Agricole group)

January 2023 – September 2025

*Tech lead, card and tokenisation platform*

- Application architecture and development of the Java 21 / Spring Boot 3 microservices for the card lifecycle (issuing, activation, blocking, PIN) and Apple Pay / Google Pay tokenisation, in hexagonal architecture.
- Tokenisation performance: network latency traced end to end in Dynatrace, microservices merged to remove intermediate calls, request-reply over Kafka replaced by direct HTTP calls, PostgreSQL indexes reworked.
- Balance check between the bank, the card processor and Visa: investigated the bank’s response times and the use of stand-in authorisation.
- Data-flow architecture: Kafka event-driven design, and the trade-off between synchronous calls and Kafka to meet Visa's response times; REST / OpenAPI contracts with the card processor and the core banking system.
- Latency between GCP and on-premises: interconnect gateway and direct DNS resolution checked with the cloud network team.
- Migration from Spring Boot 2 to 3 and GraalVM Native Image; GitLab and ArgoCD CI/CD on GKE, SonarQube, CVE tracking, Datadog and Dynatrace observability; Clean Code and TDD / BDD spread through code review.

*Java 21, Spring Boot 3, Kafka Confluent and Schema Registry, REST / OpenAPI, GCP / GKE and on-premises, ArgoCD, GitLab CI/CD, PostgreSQL, GraalVM, Datadog, Dynatrace*

**Société Générale** July 2021 – December 2022

*Team lead and Java developer, retail banking document management APIs*

- Architecture of the Spring Boot APIs in front of document management and of the Spring Batch jobs, within a SAFe train.
- Gatling load tests: bottlenecks found and fixed (saturated connection pool, slow queries), response times reduced; deployment automation, code review, onboarding of newcomers.

  *Java 8, Spring Boot 2, Spring Batch, MongoDB, S3-compatible object storage, Gatling, Jenkins*

**BPCE-IT** October 2019 – July 2021

*Tech lead, group private cloud portal*

- Event Sourcing / CQRS application architecture and the portal's state machine; stateless Spring Boot microservices built with TDD.
- Automated virtual-machine provisioning flows and multi-provider management over RabbitMQ; attribute-based access control (ABAC), cost chargeback.

  *Java 8, Spring Boot, Event Sourcing / CQRS, RabbitMQ, MongoDB, PostgreSQL, Jenkins*

**Bpifrance** February 2018 – September 2019

*Development team lead, customer portal for financing products*

- Application architecture of the new customer portal in Clean Architecture and of the REST exchanges; technical leadership and code review; REST services with TDD, Jenkins CI, OpenShift deployment.

  *Java 8 / 10, Spring Boot 2, JAX-RS, JPA, Angular, Oracle, Jenkins, OpenShift*

**Natixis Wealth Management** September 2015 – February 2018

*Java developer and team lead, KYC, customer risk rating and identity federation*

- Private-bank applications in hexagonal architecture (customer risk rating under AML and ACPR requirements, KYC completeness, mobile customer area); identity federation with a SAML identity provider (OpenSAML); life-insurance application services.

  *Java 7 / 8, Spring / Spring Boot, JBoss, SQL Server, OpenSAML, Jenkins*

## EARLIER EXPERIENCE

Sopra Steria, for RATP and Île-de-France Mobilités (February 2012 – August 2015): Java / JEE developer. Real-time passenger information flows (OSGi / Karaf, Apache Camel, CXF, ActiveMQ), traffic control interface, geographic reference data (Hibernate, PostgreSQL).

## PERSONAL PROJECTS

**IAMINE Compta (2026):** double-entry accounting on top of a banking API; PostgreSQL ledger whose balance is enforced by the database, chained immutable entries, bank ingestion published to Kafka with one idempotency key per transaction (Quarkus / GraalVM). Replaying a full fiscal year gives the same income statement as the accounting firm's.

**Liam Medical (2025 – 2026): medical practice software designed alone and delivered to a practice in March 2026 (Java 21, Spring Boot 3, PostgreSQL); the AI assistance runs locally, so a patient's record never goes to an online service.**

## EDUCATION AND LANGUAGES

Master's degree in business IT (MIAGE), Université Nice Sophia Antipolis (2012). French (native), professional English, Arabic.